MICROBIOLOGY · MCQ SET

# Blood Flukes (Schistosomes)

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Microbiology GUE Block · MEDucation by Mattin Majid · Practice MCQs

20 questions, mixed difficulty, 9 case-based. Attempt every question before checking the answer key on the last page.

| 4 | 10 | 6 |
|---|---|---|
| Easy | Medium | Hard |

1. EASY

**What type of organism causes schistosomiasis?**

A. A cestode (tapeworm)
B. A nematode (roundworm)
C. A dioecious trematode (fluke)
D. A protozoan
E. A monoecious (hermaphroditic) trematode

2. EASY

**Who first described schistosomes, and in which organ?**

A. Robert Koch, in the lungs
B. Theodor Bilharz, in the mesenteric veins
C. Louis Pasteur, in the liver
D. Patrick Manson, in the bladder
E. Alphonse Laveran, in the blood

3. EASY

**What is the infective stage of Schistosoma for humans?**

A. Egg
B. Miracidium
C. Sporocyst
D. Cercaria
E. Adult worm

4. EASY

**What is the drug of choice for treating all three major Schistosoma species?**

A. Albendazole
B. Metronidazole
C. Praziquantel
D. Ivermectin
E. Niclosamide

5. MEDIUM

**What structure holds the female worm within the male during copulation?**

A. The oral sucker
B. The acetabulum
C. The gynecophoric canal
D. The ventral sucker alone
E. The vitelline duct

6. MEDIUM CASE

A traveler wades in a freshwater lake in an endemic region. Within hours he develops transient itching with petechial papules exactly where his skin contacted the water.

**What is the most likely diagnosis?**

A. Cutaneous larva migrans
B. Cercarial dermatitis (swimmer's itch)
C. Katayama fever
D. Contact dermatitis from a plant
E. Bacterial cellulitis

7. MEDIUM CASE

Two to three weeks after freshwater exposure in an endemic area, a patient develops fever, chills, diarrhea, urticaria, lymphadenopathy and hepatosplenomegaly.

**What is the most likely diagnosis?**

A. Typhoid fever
B. Katayama fever
C. Cercarial dermatitis
D. Viral hepatitis A
E. Malaria

**8. MEDIUM**

**Which intermediate host snail genus is paired correctly with its Schistosoma species?**

A. S. mansoni — Oncomelania
B. S. haematobium — Biomphalaria
C. S. japonicum — Oncomelania
D. S. haematobium — Biomphalaria
E. S. mansoni — Bulinus

**9. MEDIUM**

**Eggs of S. mansoni and S. japonicum reach the outside environment mainly via which route, in contrast to S. haematobium?**

A. Sputum, vs. feces for S. haematobium
B. Feces, vs. urine for S. haematobium
C. Urine, vs. feces for S. haematobium
D. Blood draw only, for all species
E. Skin shedding, vs. feces for S. haematobium

**10. MEDIUM CASE**

A patient with chronic S. mansoni infection develops progressive hepatomegaly, splenomegaly and ascites, with liver synthetic function relatively preserved.

**What is driving this pathology?**

A. Direct toxin release by the adult worms
B. Granulomatous reaction to eggs carried to the liver, causing fibrosis and portal hypertension
C. A co-existing viral hepatitis infection
D. Bacterial translocation from the gut
E. Direct liver invasion by cercariae

**11. MEDIUM**

**Which of the following is the recognized long-term malignant complication of chronic S. haematobium infection?**

A. Hepatocellular carcinoma
B. Cholangiocarcinoma
C. Squamous cell carcinoma of the bladder
D. Renal cell carcinoma
E. Colorectal adenocarcinoma

**12. MEDIUM CASE**

A patient with suspected urinary schistosomiasis needs histopathological confirmation.

**Which biopsy site is most appropriate to confirm this diagnosis histologically?**

A. Liver biopsy only
B. Bladder mucosal biopsy
C. Skin biopsy at the entry site
D. Bone marrow biopsy
E. Lymph node biopsy

**13. HARD**

**Granulomas in schistosomiasis form primarily in response to what stimulus?**

A. Adult worm cuticle antigens
B. Cercarial surface antigens
C. Antigens secreted by eggs
D. Snail-derived antigens
E. Miracidial locomotor cilia

**14. HARD CASE**

Cystoscopy in a patient with chronic urinary schistosomiasis shows a granular, sandy-appearing patch on the bladder mucosa.

**What is the most likely explanation for the 'sandy patch' appearance?**

A. Calcium deposition unrelated to infection
B. Generalized hyperplasia and fibrosis of the vesical mucosa from chronic egg deposition
C. A separate fungal infection of the bladder
D. Urothelial carcinoma in situ
E. Tuberculous cystitis

**15. HARD CASE**

A patient with long-standing S. haematobium infection develops progressive obstructive hyperplasia of the ureters and urethra.

**Which complication should be specifically evaluated for, given the mechanism described?**

A. Hydroureter/hydronephrosis from obstructive hyperplasia
B. Hepatocellular carcinoma
C. Pulmonary embolism
D. Peripheral neuropathy
E. Adrenal insufficiency

**16. HARD**

**A rectal mucosal biopsy in a patient from an endemic area shows egg clusters in the lamina propria with characteristic lateral spines. Which two species does rectal biopsy specifically help diagnose?**

A. S. haematobium and S. mekongi
B. S. mansoni and S. japonicum
C. S. haematobium only
D. S. japonicum only
E. All five Schistosoma species equally

**17. HARD**

**Besides the bladder/intestine and liver, which other organ systems are stated to be sites of significant disease in schistosomiasis?**

A. Lungs and central nervous system (spinal cord)
B. Thyroid and parathyroid
C. Pancreas and gallbladder only
D. Bone marrow only
E. Skin only, beyond the entry site

**18. HARD CASE**

A child in an endemic area has chronic, longstanding schistosomiasis.

**What chronic hematologic complication is this child at particular risk for?**

A. Polycythemia
B. Anemia
C. Thrombocytosis
D. Leukemoid reaction
E. Methemoglobinemia

**19. HARD CASE**

Public health officials plan a schistosomiasis control program for a high-prevalence rural community.

**Which single prevention measure most directly interrupts the life cycle at the intermediate host stage?**

A. Praziquantel mass treatment of infected persons
B. Eradication of water snails
C. Avoiding freshwater contact
D. Boiling drinking water
E. Vaccination

**20. HARD**

**A biopsy shows parasite eggs in the lamina propria with visible miracidial nuclei and characteristic terminal spines. From which organ was this biopsy most likely taken, and which species is implicated?**

A. Rectum; S. mansoni
B. Bladder; S. haematobium
C. Liver; S. japonicum
D. Lung; S. mekongi
E. Skin; cercarial stage

# ANSWER KEY & RATIONALE

*Don't peek until you've committed to an answer on every question.*

| Q | Ans | Why |
|---|---|---|
| 1 | C | Schistosomes are dioecious (sexes separate) trematodes, unlike most flukes which are hermaphroditic. |
| 2 | B | Theodor Bilharz first observed the worm in the mesenteric veins of an Egyptian in Cairo in 1851. |
| 3 | D | Cercariae are the infective stage, penetrating unbroken human skin. |
| 4 | C | Praziquantel is the treatment of choice for all three major species. |
| 5 | C | The gynecophoric canal is the ventral groove formed by the male's folded body that holds the female. |
| 6 | B | Transient itching with petechial lesions at the water-contact site, appearing within hours, is cercarial dermatitis (swimmer's itch). |
| 7 | B | Fever, chills, diarrhea, urticaria, lymphadenopathy and hepatosplenomegaly 2–3 weeks after exposure is Katayama fever, the acute stage. |
| 8 | C | S. japonicum's intermediate host is Oncomelania; S. mansoni uses Biomphalaria and S. haematobium uses Bulinus. |
| 9 | B | S. mansoni and S. japonicum eggs are shed in feces; S. haematobium eggs are shed in urine, reflecting their different habitats. |
| 10 | B | Hepatic fibrosis and portal hypertension in schistosomiasis result from the granulomatous reaction to eggs carried to the liver, not direct worm toxicity. |
| 11 | C | Chronic S. haematobium infection is linked to squamous cell carcinoma of the bladder via chronic granulomatous inflammation and fibrosis. |
| 12 | B | Bladder mucosal biopsy is used to demonstrate S. haematobium eggs; rectal biopsy is used for S. mansoni and S. japonicum. |
| 13 | C | Granulomas form in response to antigens secreted by the eggs, not the adult worms or cercariae. |
| 14 | B | The sandy patch reflects generalized hyperplasia and fibrosis of the vesical mucosa from chronic egg deposition and the resulting granulomatous response. |
| 15 | A | Obstructive hyperplasia of the ureters/urethra leads to hydroureter and hydronephrosis, which should be evaluated for (e.g. by ultrasound). |
| 16 | B | Rectal mucosal biopsy demonstrates eggs of S. mansoni and S. japonicum; bladder biopsy is used for S. haematobium. |
| 17 | A | The lecture specifically notes significant disease may occur in the lungs and central nervous system (spinal cord), with possible genital organ involvement. |
| 18 | B | Chronic schistosomiasis, especially in children, is stated to lead to anemia. |
| 19 | B | Eradication of the intermediate hosts (water snails) directly interrupts the life cycle at the snail stage, one of the listed prophylactic measures alongside treating infected persons and avoiding contaminated water. |
| 20 | B | Bladder mucosal biopsy showing egg clusters, miracidial nuclei and characteristic terminal spines is used to diagnose S. haematobium. |